# Amplicon library workflow: two-step PCR or one-pot PCR

both routes give the same barcoded molecule and share QC, pooling, purification and Evoprep · HuangLab LIMS stages 1 → 3a → 3b → 4a

**Stage 1** — Specimen# / Culture#

**Stage 3a** — extraction + Qubit · MolWork#

**Stage 3b** — PCR + gel + Qubit · por_reaction row

**Stage 4a** — Nanopore experiment · Nanopore#

**Specimen registered in LIMS**
soil / environmental → Specimen# | fungal strain → Culture#

**LIMS register**
`/lims-register`
Stage 1 → Specimen# (soil / env.) or Culture# (strain)

**Soil DNA extraction**
SuperSoil protocol

**Fungal DNA extraction**
modified BOMB v9.0

**DNA extraction**

**DNA template → MolWork#**
soil crude extract (SuperSoil) | pure fungal extraction (BOMB v9)
MolWork# minted after quantification



**Qubit Flex**
quantify the extract (Stage 3a)

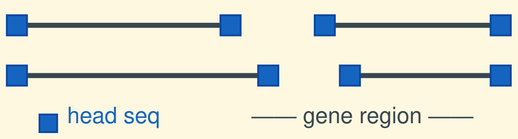
**LIMS register**
`/lims-register`
Stage 3a → MolWork# minted after quantification

choose a route

## Route A · two-step PCR

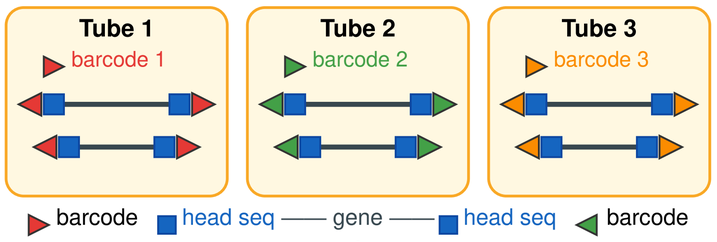
**1st PCR (gene regions)**

1st PCR products (per sample)




**2nd PCR (barcoding)**
gel check #1: good amplicons only

2nd PCR products: one tube per sample, one barcode per tube




**2-step PCR protocol**
mixtures + cycling

## Route B · one-pot PCR

**Set up one-pot PCR**

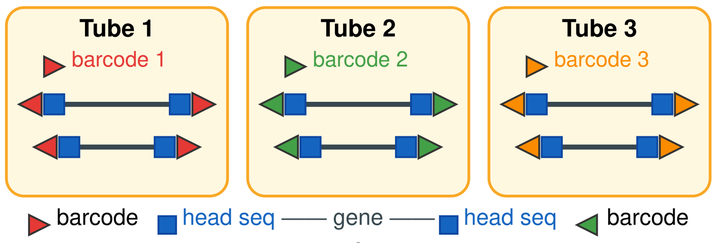
One reaction tube (per sample)

**gene-region primers** (head-tailed, 0.1 µM each) + **barcode primers** (barcode + head seq, 0.3 µM each)

50 µL: KAPA HiFi 2X + both primer sets + 3 µL template

**One-pot PCR**
phase 1: 25 cycles, 60–66 °C
phase 2: 15 cycles, 65 °C

One-pot PCR products: one tube per sample, one barcode per tube




**One-pot PCR protocol**
50 µL mix + 25/15 cycles

same molecule from either route



**Gel check → Qubit Flex (Stage 3b)**
confirm barcoded product, quantify → 10 ng/µL



**LIMS register**
`/lims-register`
Stage 3b → pcr_reaction: PCR_<Ta> / one_pot_<Ta>

**Pooling (equal molar)**

**Pool all barcoded samples**
equal molar of every sample



**Pooled sample**

**Purification (bead clean-up)**
PCR sample : bead = 1 : 1



**Purified pooled sample**

**Size selection 1X**
bead purification protocol

**Ready to be sequenced**
library construction using Evoprep

**EvoPrep library prep**
KAPA EvoPrep NBD114.96

**LIMS register**
`/lims-register`
Stage 4a → Nanopore#: experiment, samples, NanoAmp

**Legend**

- Head seq (universal tail on the gene-region primers)
- Barcode (sample-specific, on the barcode primers)
- Gene region (target amplicon; colour = different sample)

QR codes (left) open the lab's self-hosted protocol PDF for that step
left bands = LIMS stages; /lims-register at every stage; dashed = QC / info flow
Route A: extra gel check after the 1st PCR · Route B: none